



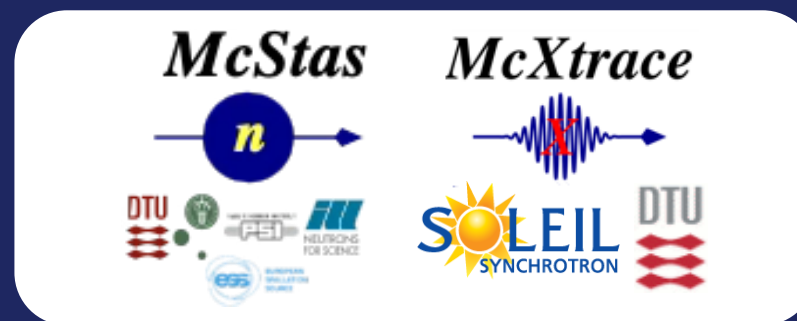
A COMMON INTRODUCTION

McCode-vibing

Experiences using standard LLM models for writing instruments etc.

McStas — neutrons

McXtrace — X-rays



PaNRAID School · supported by PEPR DIADEM



Examples of vibe-coding with McCode

1

“Build me a powder diffractometer ”

One codebase family, two physics engines

2

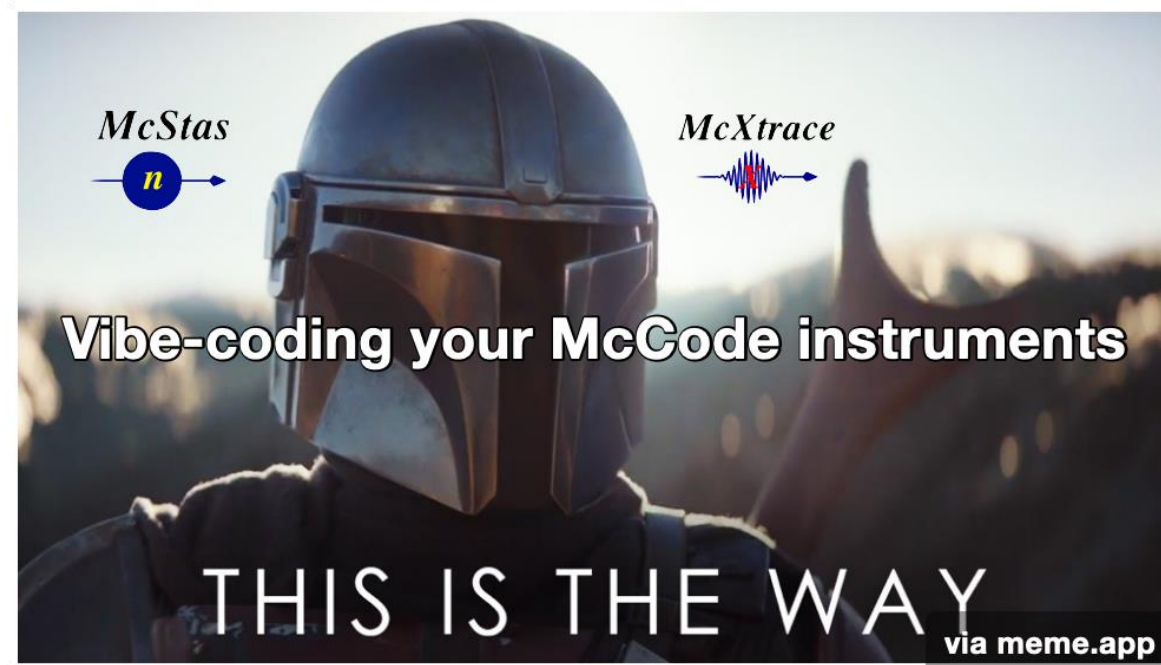
“Use information on the web, draft me an instr”

Instruments, components, and the ray state

3

Full use-case from Anders Kästner, PSI

Where the two probes really differ





“Build me a powder diffractometer ”

Prompt:

Based on available information in <https://mcstas.org> and <https://github.com/mccode-dev/McCode/wiki>, please put together the code for a diffractometer at a reactor source.

Here are the main characteristics

1. Input parameter lambda (wavelength), used to calculate the relevant Bragg-angle for Monochromator. Default to 2.567 Angstrom
2. Source with 3 Maxwellians
3. 10 m flight path with 4 cm wide, 8 cm tall neutron guide
4. Single slab PG Monochromator, rotated according to calculated Bragg angle
5. 2 m distance to Sample (further rotated to be in the scattered beam from the mono
6. Powder sample radius 4 mm, 70 mm tall cylinder, material $\text{Na}_2\text{Ca}_3\text{Al}_2\text{F}_{14}$
7. Banana-shaped detector (cylindrical cut) 10 cm tall, radius 1.5 m from sample, to record 10-180 degrees on the "high-resolution" side of the sample





“Build me a powder diffractometer” – response:

Reactor_PowderDiffractometer.instr is ready. It builds a constant-wavelength reactor powder diffractometer in the style of PSI's DMC / ILL's D2B/D20:

- **lambda** (default 2.567 Å) drives the Bragg-angle calculation in INITIALIZE: $\theta_M = \text{asin}(\text{mono_Q} \cdot \lambda / (4\pi))$, using $\text{mono_Q} = 1.8734 \text{ Å}^{-1}$ for PG(002). At the default wavelength this gives $\theta_M \approx 22.5^\circ$, $2\theta_M$ take-off $\approx 45^\circ$.
- **Source:** Source_Maxwell_3 with the canonical three-Maxwellian parameter set from McStas's own source examples (T1=216.8 K, T2=33.9 K, T3=16.7 K with matching intensities) as a representative reactor-moderator spectrum. The simulated band is narrowed to $\lambda \pm \Delta\lambda$ for statistics efficiency (a standard trick for fixed-wavelength instruments — widen $\Delta\lambda$ to see the full spectrum).
- **Guide:** 10 m, 4 cm × 8 cm, m=2 supermirror.
- **Monochromator:** single-slab Monochromator_flat (PG mosaic, 5×5 cm), placed on an Arm rotated by the computed θ_M ; a second arm downstream picks up the full $2 \cdot \theta_M$ deflection.
- **Sample:** PowderN with Na2Ca3Al2F14.laz (the standard McStas NAC reflection data file — same one used in the real DMC instrument file), radius 4 mm, height 70 mm, wrapped in a SPLIT for better detector statistics.
- **Detector:** Monitor_nD in "banana, theta" mode, radius 1.5 m, height 10 cm, covering 10–180°, placed on the sample arm and rolled 180° about the beam axis — the same construction used in the real PSI_DMC instrument to put the bank on the focusing/high-resolution side matching the monochromator's take-off direction.

Everything is exposed as instrument parameters (guide length/size, monochromator size/mosaic, distances, sample and detector geometry) so you can retune it.

I validated this against the real thing rather than just from memory: I pulled the actual .comp source files for every component used (Source_Maxwell_3, Guide, Arm, Monochromator_flat, PowderN, Monitor_nD, PSD_monitor, L_monitor, Progress_bar) from the McCode GitHub repo, and parsed the instrument file with mccode-antlr (the McStas grammar/compiler front-end) against those real definitions. It resolved all 14 components and every parameter name cleanly with no errors; I also confirmed the checker actually catches mistakes by deliberately introducing a typo'd parameter and an undefined component reference, both of which it flagged. Full C-code generation itself was blocked only by a sandboxed network path (github.com was reachable via raw.githubusercontent.com but not directly), which is unrelated to the instrument's own correctness. I also cross-checked the file against McStas's real PSI_DMC.instr example (same default wavelength, same detector-radius convention, same rotation idiom for the "high-resolution side"), fetched live from the repo. Sources: ...

Example 1



“Build me a powder diffractometer” – response:

```

*****
* McStas instrument: Reactor_PowderDiffractometer
*
* %Identification
* Written by: Claude (Anthropic), for Peter
* Date: 2026-08-23
* Origin: generic reactor source
* %INSTRUMENT_SITE: Templates
*
* %Description
* Generic constant-wavelength (CW) powder neutron diffractometer for
* reactor source, in the style of instruments such as PSI's DMC or
* D2B/D20. A triple-Maxwellian reactor source feeds a 10 m straight
* a single-slab pyrolytic graphite (PG(002)) flat monochromator sele
* wavelength and defines the Bragg take-off angle, and a cylindrical
* "banana" detector bank records the powder diffraction pattern from
* sample.
*
* The monochromator is rotated automatically from the requested wave
* 'lambda' using Bragg's law (theta_M = asin(mono_Q*lambda/(4*PI)))
* sample and detector bank are placed on the arm that carries this
* rotation, i.e. in the beam reflected by the monochromator, on the
* rotational side as the monochromator take-off ("high-resolution"
* focusing side, following the same convention used in the McStas P
* example instrument).
*
* The source spectrum uses the canonical three-Maxwellian parameter
* (T1=216.8 K, T2=33.9 K, T3=16.7 K with matching intensities) that
* with McStas' own Source_Maxwell_3 test/example instruments, as a
* physically reasonable stand-in for a real reactor moderator spectr
* Because this is a fixed-wavelength (CW) instrument, the simulated
* band is narrowed to +/- dlambda_pct % around lambda for statistics
* efficiency; widen 'dlambda_pct' (or edit Lmin/Lmax) to inspect the
* source spectrum.
*
* The powder sample (Na2Ca3Al2F14, "NAC") has many active reflector
* this wavelength, so PowderN's 'd_phi' vertical-focusing is used to
* restrict the simulated Debye-Scherrer cones to the detector height
* is computed automatically from det_yheight/det_radius), and 'tth_sign' is
* fixed to steer scattering onto the same side the detector bank actually
* occupies, instead of wasting half the statistics on the far side. A
* SPLIT of 58 after the sample is a good default multiplicity for this
* sample/wavelength combination.
*
* %Example: lambda=2.567 Detector: Detector_I=1e3
*
* %Parameters
* lambda: [AA]      Wavelength selected by the monochromator (sets Bragg
* mono_Q: [AA-1]    Scattering vector of the monochromator reflection (
* dlambda_pct: [%]  Half-width of the simulated source band around lamb
* L_source_guide: [m] Distance from source (moderator face) to guide ent
* L_guide: [m]      Guide length

```

```

COMPONENT mono_axis = Arm()
AT (0, 0, L_guide_mono) RELATIVE PREVIOUS

COMPONENT mono_rotation = Arm()
AT (0, 0, 0) RELATIVE mono_axis
ROTATED (0, theta_M, 0) RELATIVE mono_axis

COMPONENT mono = Monochromator_flat(
  zwidth = mono_zwidth, yheight = mono_yheight,
  mosaich = mono_mosaic, mosaicv = mono_mosaic,
  r0 = mono_r0, Q = mono_Q)
AT (0, 0, 0) RELATIVE mono_rotation

/* Output arm: carries the full 2*theta_M deflection of the reflected beam */
COMPONENT mono_out = Arm()
AT (0, 0, 0) RELATIVE mono_rotation
ROTATED (0, theta_M, 0) RELATIVE mono_rotation

* -----
* Sample position, 2 m down the beam reflected by the monochromator
* (i.e. rotated by 2*theta_M into the scattered beam).
* -----

COMPONENT sample_arm = Arm()
AT (0, 0, L_mono_sample) RELATIVE mono_out

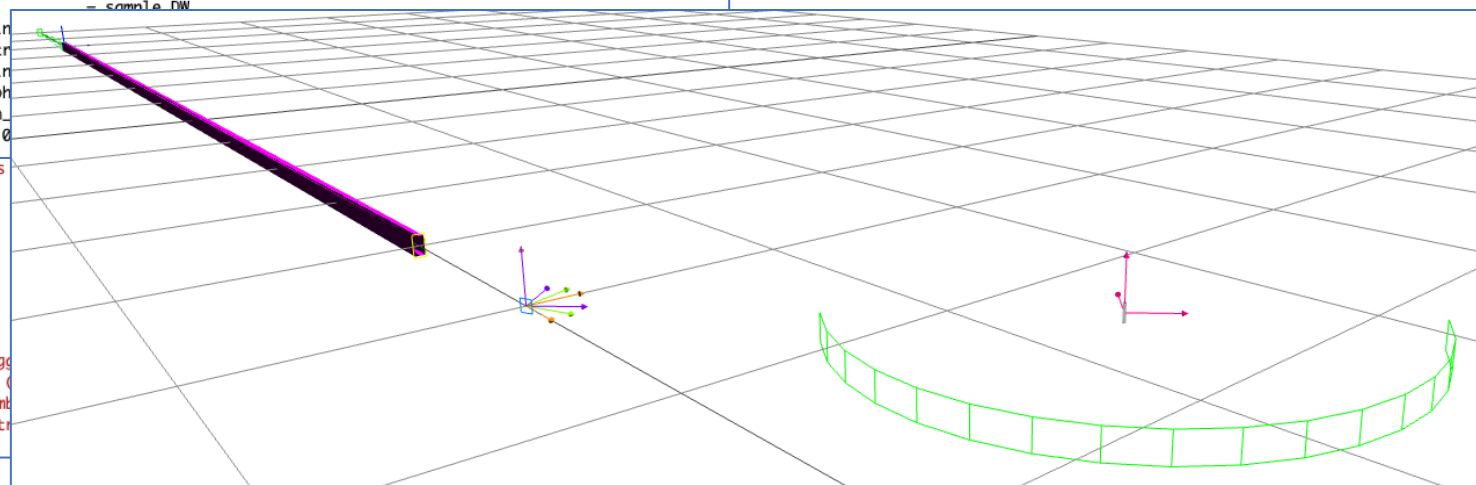
SPLIT sample_split COMPONENT sample = PowderNC
  radius      = sample_radius,
  yheight     = sample_yheight,
  reflections  = "Na2Ca3Al2F14.laz",
  pack        = sample_pack,
  DW          = sample_DW

p_in
p_tr
p_in
d_phi
tth
AT (0, 0

```

Simulation from prompt 1 – junk!

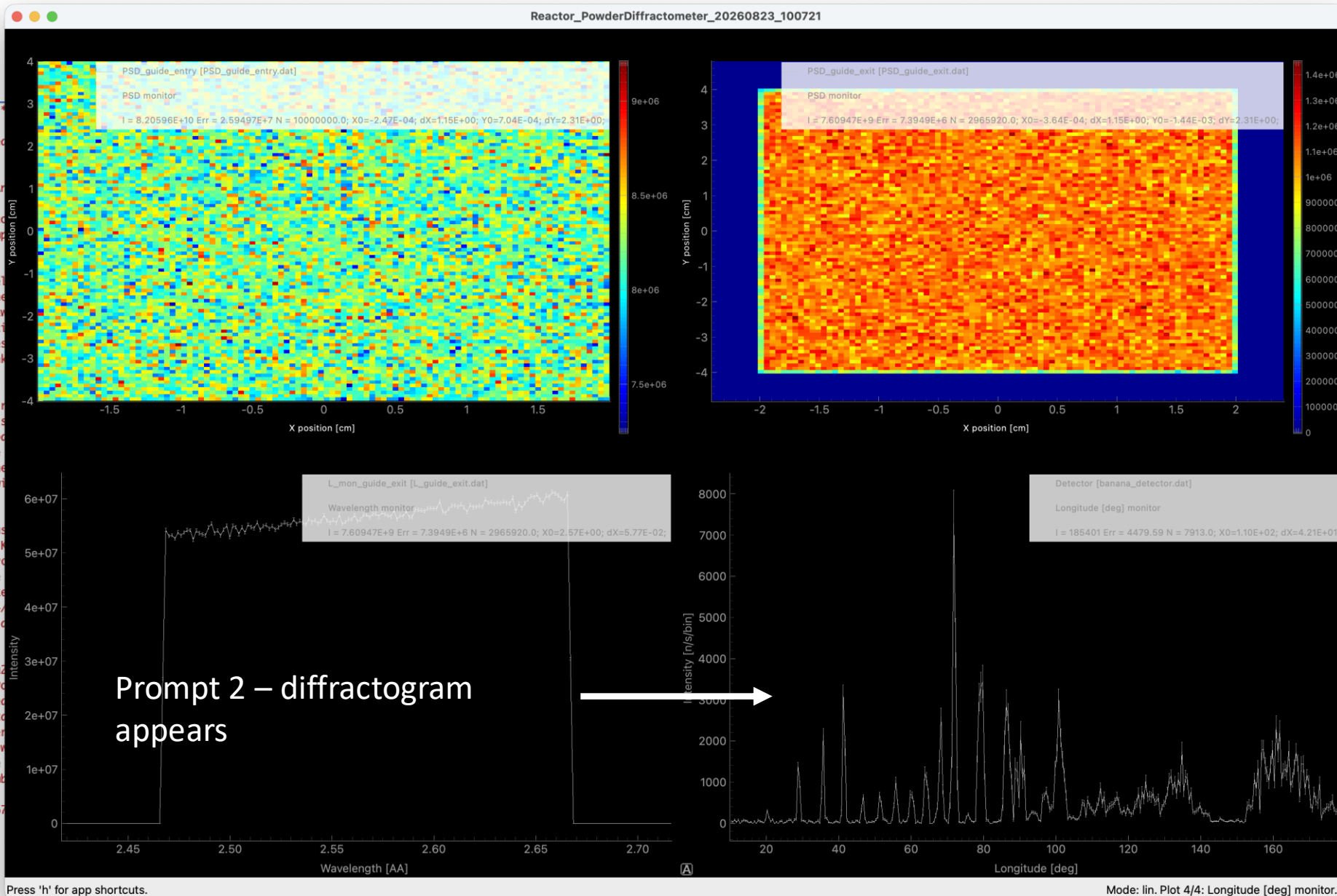
(My fault, the detector picked up direct beam from the Mono...)



Example 1



“Bu

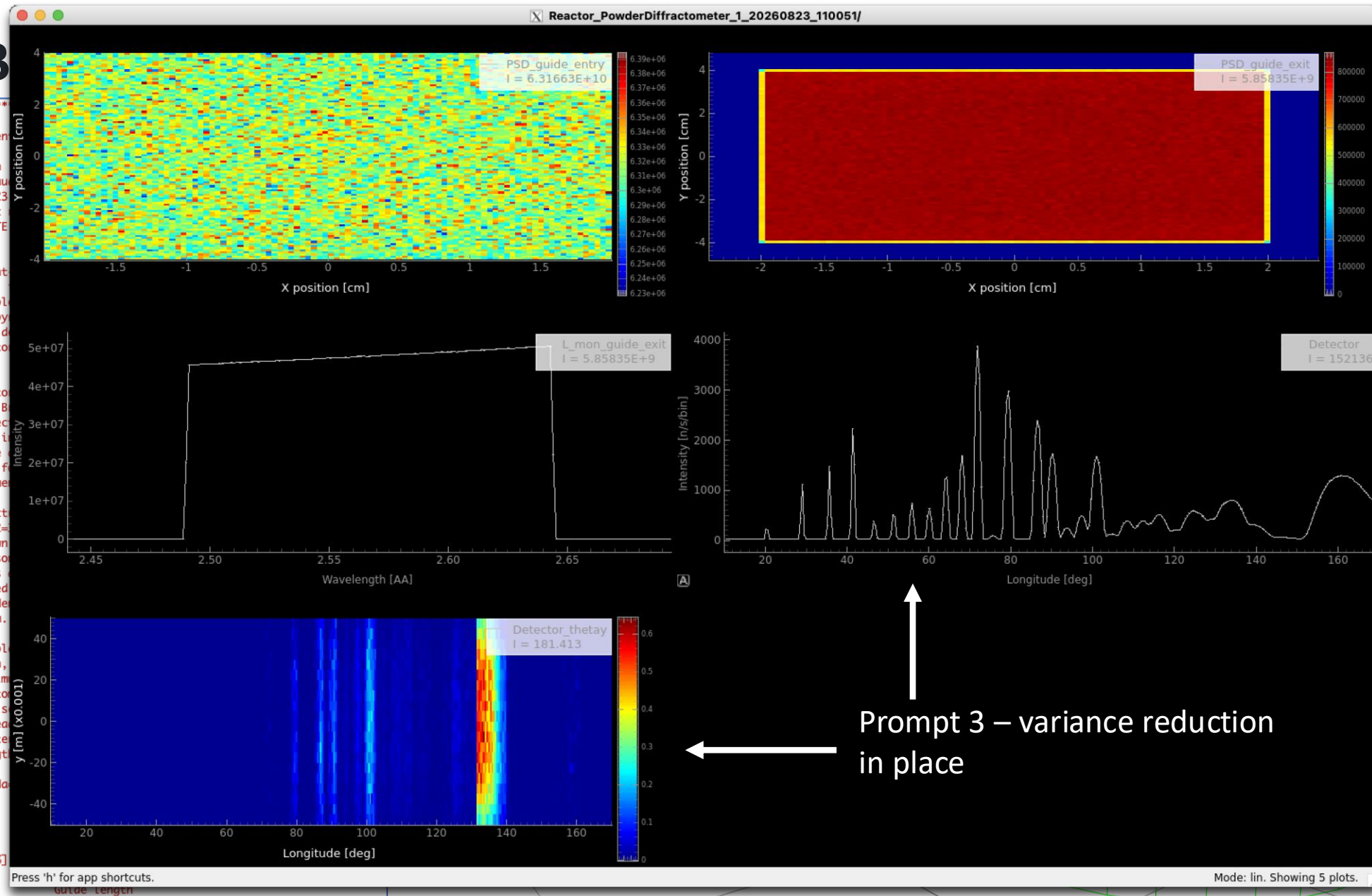


Example 1



"B"

```
*****
* McStas instrument
* %Identification
* Written by: Claude
* Date: 2026-08-23
* Origin: generic
* %INSTRUMENT_SITE
*
* %Description
* Generic constant
* reactor source,
* D2B/D20. A triple
* a single-slab py
* wavelength and d
* "banana" detector
* sample.
*
* The monochromator
* 'lambda' using B
* sample and detec
* rotation, i.e. i
* rotational side
* focusing side, f
* example instrum
*
* The source spect
* (T1=216.8 K, T2=
* with McStas' own
* physically reason
* Because this is
* band is narrowed
* efficiency; wide
* source spectrum.
*
* The powder sample
* this wavelength,
* restrict the sim
* is computed auto
* fixed to steer s
* occupies, instea
* SPLIT of 58 after
* sample/wavelength
*
* %Example: lambda
*
* %Parameters
* lambda: [AA]
* mono_Q: [AA-1]
* dlambda_pct: [%]
* L_source_guide:
* L_guide: [m]
```



Prompt 3 – variance reduction in place





Build me a model of ILL XX

Prompt 1:

Try building me an instrument that doesn't exist ready-made in McStas, ILL D17

<https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d17/>

Prompt 2:

Can you make me an ILL_D3 model? Use the information in

<https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d3/>

For the Source term (hot neutrons) I believe that either Source_gen or Source_Maxwell_3 has a listing of relevant parameters.

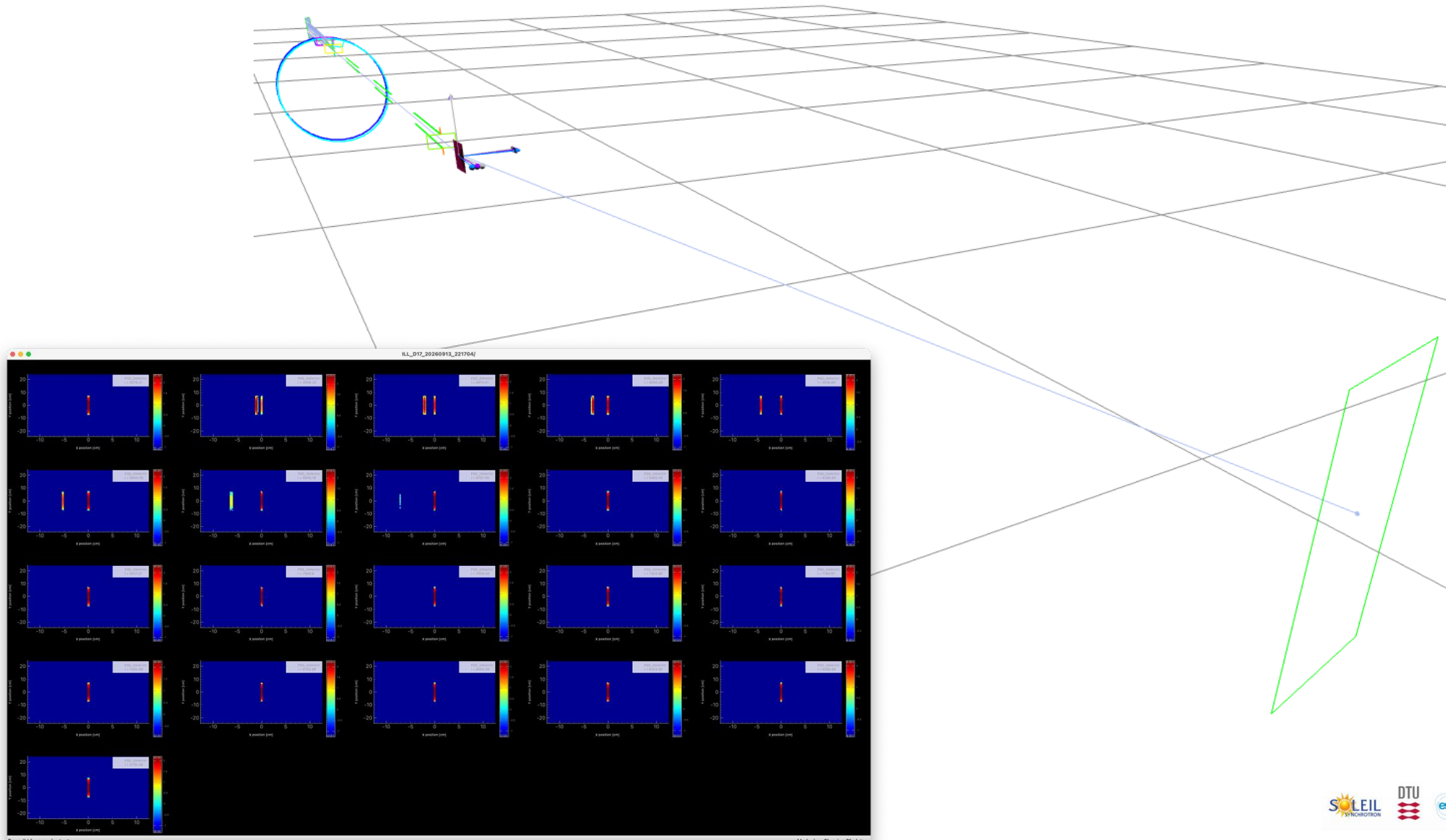
There are multiple Monochromator-options on this instrument, Monochromator_pol may be useful for the polarising mono (partly a phenomenological model), whereas the non-polarised may be simulated using either Monochromator_curved (phenomenological model) or NCrystal_sample.

If you find a design-oriented publication that would be useful for certain numbers I can probably fetch you that for round 2.



Example 2

Build me a model of ILL D17



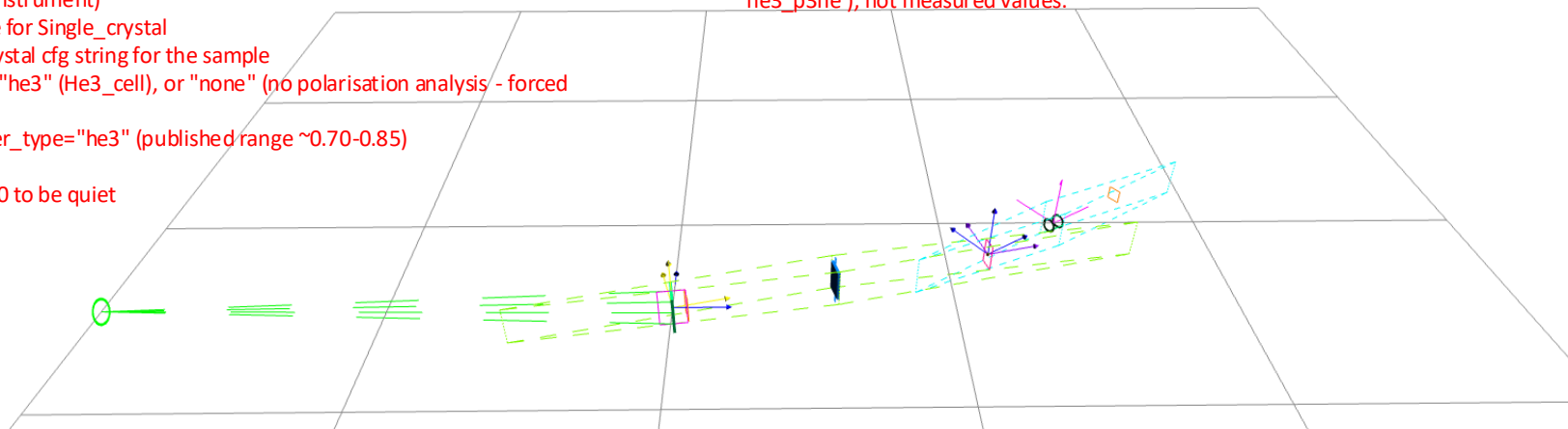


Build me a model of ILL D3

* Sources: <https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d3/>
<https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d3/characteristics/>
<https://www.ill.eu/en/for-ill-users/instruments/instruments-list/d3/how-it-works/spherical-polarimetry-with-cryopad>

* %Parameters
 * mono_type: [str] "Heusler" (polarising, high flux), "CoFe" (polarising, better res), or "Cu" (non-polarising)
 * lambda: [AA] Incident wavelength (0.42-0.843 Angs on the real instrument)
 * THETA_M: [deg] Monochromator take-off half-angle. 0 (default): derive from lambda and DM. Real range: 3.5-19 deg (i.e. $7 < 2 * THETA_M < 38$)
 * use_ncrystal: [1] Cu branch only: 0 (default) Monochromator_curved mosaic model, 1 NCrystal_sample with ncrystal_mono_cfg
 * ncrystal_mono_cfg: [str] NCrystal cfg string for the Cu monochromator, used only if use_ncrystal=1
 * mono_Rup: [1] [assumed] Polarising monochromator reflectivity for spin-up
 * mono_Rdown: [1] [assumed] Polarising monochromator reflectivity for spin-down
 * mono_mosaic: [arc min] Mosaic spread used for the Cu (Monochromator_curved) and polarising crystals
 * L1: [m] [assumed] Source-monochromator distance
 * L2: [m] [assumed] Monochromator-sample distance
 * L3: [m] [assumed] Sample-detector distance
 * flip: [1] 0: spin-up (non-spin-flip) incident beam. 1: spin flipped (spin-flip channel) - toggles an ideal Mezei flipper
 * guide_field: [T] [assumed] Vertical guide field maintaining the polarisation quantisation axis between monochromator and analyser
 * sample_engine: [str] "vanadium" (isotropic incoherent calibration sample), "sx" (Single_crystal, supply 'reflections'/UB matrix), "ncrystal" (NCrystal_sample, supply ncrystal_sample_cfg), or "none" (direct beam)
 * sample_diam: [m] Sample diameter (≤ 0.012 m on the real instrument)
 * sample_height: [m] Sample height (≤ 0.012 m on the real instrument)
 * reflections: [str] sample_engine="sx" only: reflection list file for Single_crystal
 * ncrystal_sample_cfg: [str] sample_engine="ncrystal" only: NCrystal cfg string for the sample
 * analyzer_type: [str] "ideal" (PolAnalyser_ideal, vertical axis), "he3" (He3_cell), or "none" (no polarisation analysis - forced when mono_type="Cu")
 * he3_p3he: [1] [assumed] 3He cell polarisation for analyzer_type="he3" (published range $\sim 0.70-0.85$)
 * det_2theta: [deg] Detector arm scattering angle 2theta
 * verbose: [1] Print the derived instrument configuration. 0 to be quiet

* KNOWN SIMPLIFICATIONS / THINGS NOT MODELLED (please tighten with real engineering numbers before quantitative use):
 * - CRYOPAD's zero-field spherical polarimetry (rotating nutator fields, Meissner shields) is not modelled at all; only a single-flipper, fixed-vertical-axis longitudinal polarisation analysis is implemented.
 * - The magnetic (spin-dependent) part of the sample's scattering cross section is not modelled - none of the three 'sample_engine' choices computes a magnetic structure factor, so the flipping ratio you get out of this model reflects only the instrument's own polarising/analysing efficiency (Rup/Rdown, He3 cell polarisation, flipper), not real sample physics. That would need a bespoke sample component.
 * - L1 (source-monochromator), L2 (monochromator-sample) and L3 (sample-detector) distances, the monochromator crystal size, and the guide-field strengths are NOT published on the ILL website and are set here to plausible placeholder values, flagged [assumed] in DECLARE/the component calls; only DM values, the 7-38 deg take-off range, the 0.42-0.843 Angs band, the per-crystal flux and the 12 mm sample size are taken from the published characteristics page.
 * - The hot H4 source is modelled as a single ~ 2273 K (2000 C) Maxwellian (Source_gen), not the real multi-term fit used in ILL's yellow-book source characterisation; harmonic-contamination filters (Er, Ir, Hf, Rh, In foils) are not modelled, so higher-order $\lambda/2$, $\lambda/3$ contamination is absent here.
 * - Rup/Rdown for the polarising crystals and the He3 cell polarisation are set to plausible round numbers (see 'mono_Rup'/'mono_Rdown'/'he3_p3he'), not measured values.





Vibe coding McStas

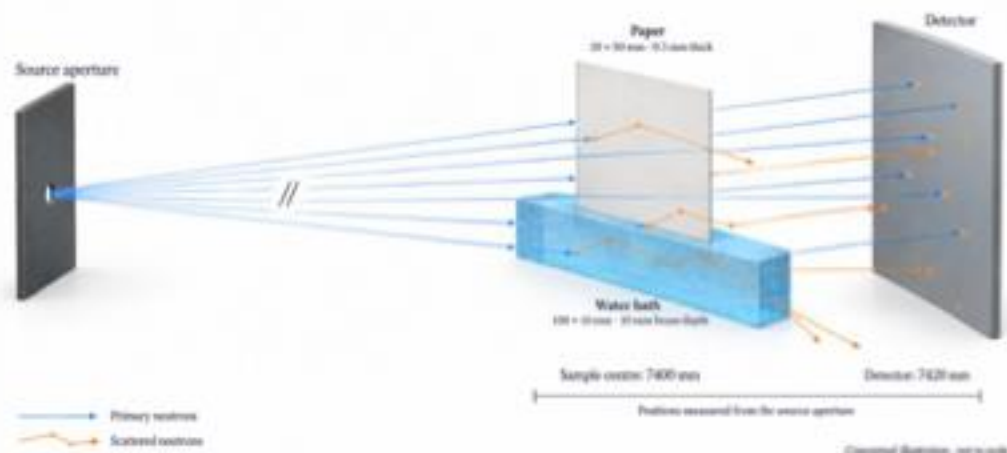
Paper in water bath

Anders Kaestner

I wanted to

- Simulate the scattering caused by a piece of paper with the edge touching a water table

Neutron scattering at a water-paper interface



- Assuming ICON as instrument
- Using McStas (I don't remember how to do it...)
- Run on my laptop

➔ Let's try vibe coding with ChatGPT plus

Prepare a bundle of the files needed to simulate the neutron scattering of a sample at 7400mm and the detector at 7420 mm from the source aperture using the mcstas model of ICON at https://mcstas.org/download/components/current/examples/PSI/PSI_ICON/PSI_ICON.html. The sample consists of two parts a water bath 10mm in the beam direction 100mm horizontal and 10mm vertical. The second part is a piece of paper 0.3mm thick, 20mm wide and 50mm tall in contact with the bath.

Sorry for the messy prompt....



Prompt breakdown

The product I want

- Prepare a bundle of the files needed to simulate the neutron scattering of a sample

Instrument conditions

- Using the mcstas model of ICON at https://mcstas.org/download/components/current/examples/PSI/PSI_ICON/PSI_ICON.html
- Sample at 7400mm and the detector at 7420 mm from the source aperture

The sample

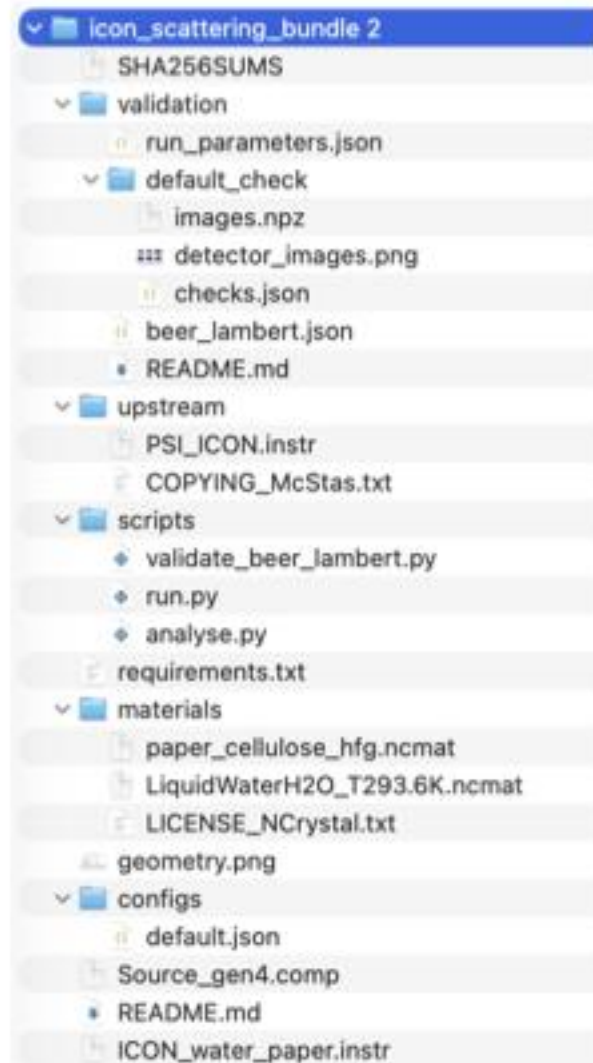
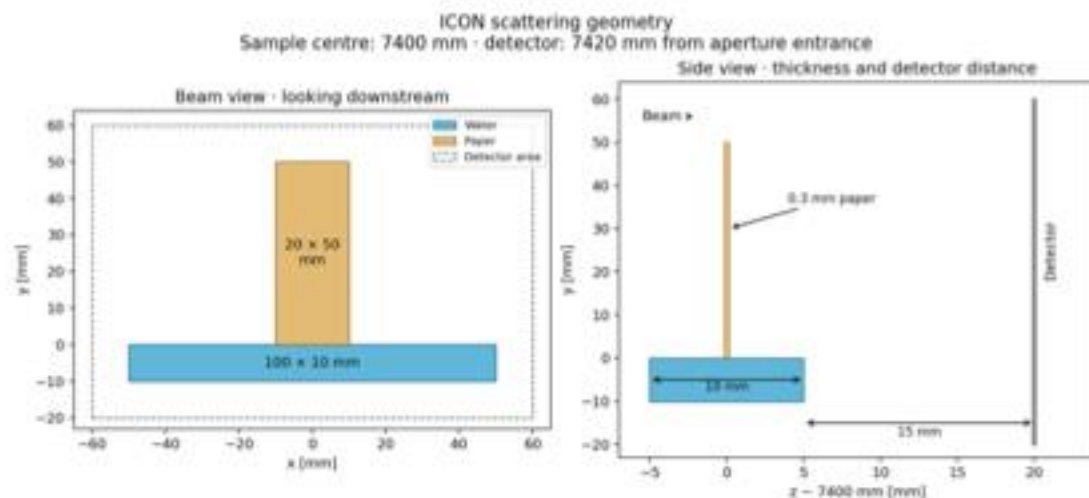
The sample consists of two parts

- a water bath 10mm in the beam direction 100mm horizontal and 10mm vertical.
- The second part is a piece of paper 0.3mm thick, 20mm wide and 50mm tall in contact with the bath.

The output

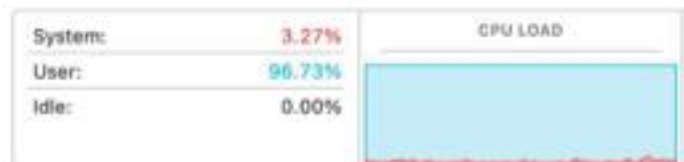
A zip file containing

- A readme.md file
- Instrument description
- Supporting instrument data
- Material descriptions
- Run-scripts



Running the simulation

- Installing McStas 3.8.4
- MacBook Pro (M1 Max, 32GB)
- Using $1e10$ neutrons for the simulation
- Image 600x400 pixels
- Run time 10min using all cores at ~100%



Example 3



Sample geometry

Instrument and detector

Simulation switches

A 'Start simulation' dialog box for the 'ICON_water_paper.instr' instrument. It contains various parameters for sample geometry, instrument and detector, and simulation switches. The 'Ray count' is set to 1000000000. The 'Start' button is highlighted in blue.

Instrument parameters: ICON_water_paper.instr

Parameter	Value	Parameter	Value	Parameter	Value
sample_z:	7.4	detector_z:	7.42	bath_width:	0.1
bath_height:	0.01	bath_depth:	0.01	paper_width:	0.02
paper_height:	0.05	paper_depth:	3003	paper_density:	0.8
immersion:	0	contact_eps:	e-09	sample_x:	0
contact_y:	0	aperture_mm:	20	extra_slit:	0
Lambda_Min:	2	Lambda_Max:	6	detector_width:	0.12
detector_height:	0.08	detector_y:	0.02	ns:	600
ny:	400	sample_enabled:	1	water_enabled:	1
paper_enabled:	1				

Simulation

Simulation/Trace/Optimize: Simulation

Output format: McCode

Autoplot: mcplot-pyqtgraph

Ray count: 1000000000

Output subdir (optional):

List mode/scan steps (optional):

Parallelisation: MPI

MPI node count: auto

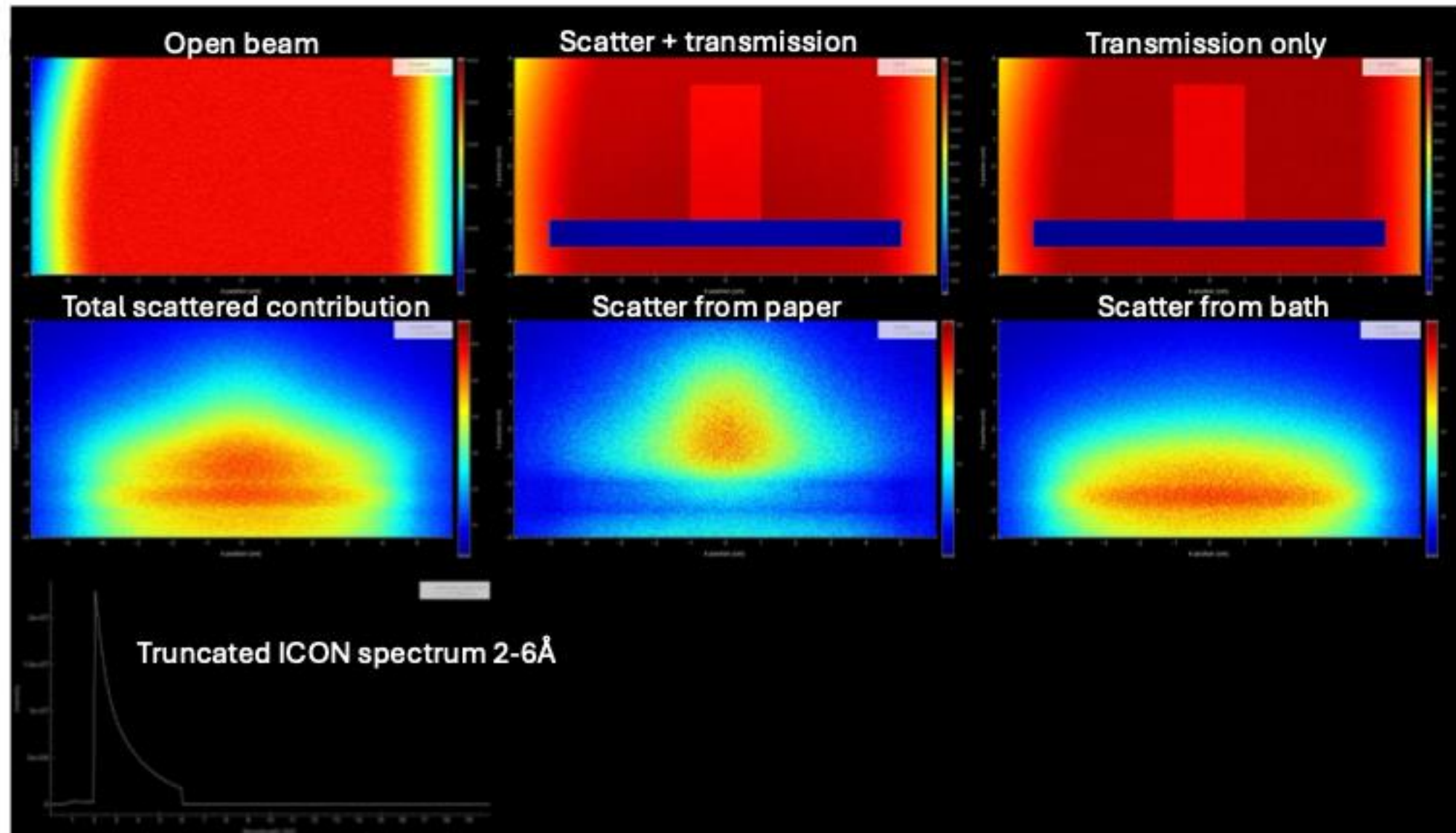
Advanced

Random seed:

Monitor_nD buffersize: 10000000

Gravity: Off

Start Cancel



Timing

Spent time from start to end

- Some minutes to formulate the prompts
- Some minutes to locate the latest McStas instrument model of ICON
- 15 min LLM time
- 10 min Download and install the latest McStas version
- 5 min Make a few test runs with less neutrons to find a decent SNR
- 10 min Make the final $1e9$ neutron simulation

In summary: less than 1h to obtain a prototype simulation using vibe coding

Conclusions

- Vibe coding seems to be a easy way to get started with McStas simulations
- It doesn't replace the full understanding what the model is actually doing
 - You have to interpret and question the model output
 - The critical thinking is still your responsibility
- This was a simple model geometry – larger models will require more time.

Prompt in
detailed fashion
and absolutely vet
the results!

Common conclusion for all 3 cases!



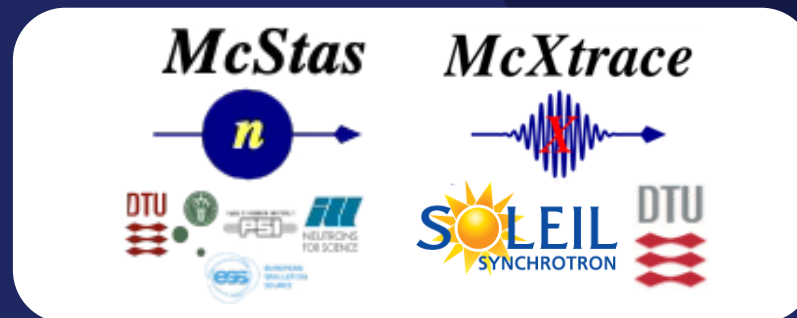


Thank you

Questions?

mcstas.org

mcxtrace.org



SUPPORTED BY



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